

Nine Dragon Stack

blow by blow

Five days, 2026-09-25 19:14 → 09-30 10:49. A Kowloon-Walled-City shard that was only meant to be concept art and a COMING SOON card, then became a playable partial shard in the real engine. This is every steering prompt, every decision, every mockup round, every phone report and every deploy, in order.

111 h

ONE SESSION, 5
DAYS, 2
COMPACTIONS

98

PROMPTS FROM
JAKE

35

DECISIONS ON
THE ASK TOOL

50

AGENT LAUNCHES

32

PRODUCTION
BUILDS SEEN

16 → 30

FPS ON A HOT
IPHONE 17 PRO



In the engine, all three real-time: the Lantern Square spawn with the Neon Jian (live 05a163d) · halfway down the Yamen Well, on a bridge looking north · the stair-street.

Live: wildshard-singleplayer.vercel.app (Nine Dragon is the EXPERIMENTAL card). Plan: docs/plans/NINE-DRAGON-STACK.md · method: docs/design/LOOK-LOOP.md · incident: docs/audits/nine-dragon-mobile-multidraw.md · asks: docs/tasks/asks/E169.md and the rest. Prompts are Jake's dictation, lightly cleaned of filler. Times are local (UTC-5). Images are from the session itself: what Jake was sent, what the agents made, what he uploaded from his phone.

09-25 19:19 · the ask

ART, A PLAN AND A TEASER CARD. NOT THE SHARD.

JAKE · 19:19

Your objective is to prepare shard 4.

- › inspiration Chongqing meets Cyberpunk 2077
- › an insanely unique shard that uses the maximum height of the chunk server and builds an insanely vertically challenged level
- › strong inspiration by the Kowloon Walled City from Hong Kong

The starting location is half way up in the sky but it has 5+ levels that all feel like the ground level. Like in Kowloon Walled City. Art style is Chinese, cyberpunk, futuristic. It's still built on the engine.

The only thing we are going to build is concept art, mega plan, mockups, the level selector coming soon assets, the from scratch clean room implementation that looks like the in game mockup of the starting spawn including the weapon of choice ... **We don't implement the shard yet.** No worktree, no branch, all on main.

Claimed **E169**. Three shards existed: Driftwood Isle (low-poly toon), Nalati Grasslands (painterly), Pine Hollow (photoreal PBR). The chunk spec was 500 × 500 m and 200 m tall. Three game browsers were already open machine-wide (the cap), so the first mockups were drawn on an old HUD capture. That mistake comes back at 19:49.

JAKE · 19:25

We can break and modify the spec: 500 m high, 250 above and 250 below. Modify the spec of the shards for this one to make it absolutely epic. And it's fine to do 500×500×500 long term.



concept 01 · the stack: a 500 m cube of city on a slab above the clouds



concept 02 · the nine strata, the Sump at -250 m to the Crown at +250 m



concept 08 · the weapon: the Neon Jian and the Fei Zhua grapple claw



concept 03 · Lantern Square, the spawn at +125 m



concept 04 · the Yamen Well: vertigo down the shaft



concept 09 · the cast

JAKE · 19:27

How are you going to do a fully custom art style for this? Do research in what would be the most insanely epic and wow.

Two codex agents ran in parallel: nine concept images and four in-game spawn compositions. A research agent wrote up candidate styles, and a style agent drew one spawn frame in six of them. The plan draft and the concept art were pushed as **ede7e60f** at 19:36. The name came from the research: 九龍, "nine dragons", which is what Kowloon means. So: **Nine Dragon Stack** (九龍疊城), nine street strata with about 12 floors of housing between each pair.



round 3 · six styles on one spawn: ink & neon · gongbi blue-green · Chungking Express · shadow theatre · neon woodblock · neon-noir realism



round 4 · the recommended custom style, 界画霓虹 Jiehua Neon: spawn · gold on indigo · down the Well · dusk

19:48 · “you’re using stale mockups of the HUD”

FRESH CAPTURES, THEN THE FIRST FOUR DECISIONS

- JAKE · 19:48

What choices are there for me to make? Show me the mockups. Then ask me any questions or decisions.
- JAKE · 19:49

You’re using stale data, or stale mockups of the HUD. The HUD has changed. I think you’re using like a broken Nalati HUD. So please take fresh screenshots of the HUD and the three shards and use that as your basis for the mockups. Just regenerate the mockups with the latest HUDs.
- JAKE · 19:50

Driftwood, the one that’s not early access and not experimental, has the baseline HUD; Pine Hollow and Nalati have shard-specific custom HUD stuff. If it’s possible within the mega plan to just use the baseline HUD, that is the absolute cleanest way of doing it.
- JAKE · 19:51

You can use the Ask User tool to ask me questions ... bonus points for a nice textual description above it, and then showing the mockups again with the minimal scroll bar on Claude iOS.



live capture 19:51 · Driftwood, the baseline HUD

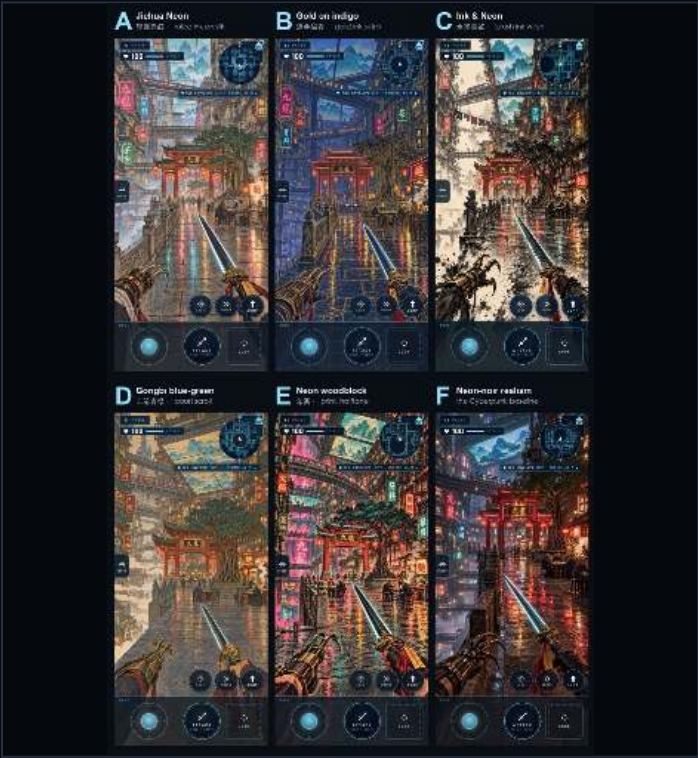


live · Nalati



live · Pine Hollow

The live build was [25166fa-muho86w1](#), our own latest commit, so these captures show today's HUD at iPhone 16 Pro size (402×874 at 3×, touch, phone tier, muted, on the GPU). All nine style and composition frames were re-rendered on Driftwood's baseline HUD and laid out as boards sized to one phone screen.



board 1 · six styles, redone on the live baseline HUD



board 2 · four spawn views in Jiehua Neon: A paifang · B Well edge · C stair-street · D down the Well

DECISION · 21:03

Which art style should Nine Dragon Stack have? (board 1)

A Jiehua Neon (Rec.)

B Gold on indigo

C Ink & Neon

F Neon-noir realism

DECISION · 21:03

Which view is the spawn and the coming-soon hero image? (board 2)

A The paifang (Rec.)

B The Well edge

C The stair-street

D Down the Well

DECISION · 21:03

The weapon of choice for this shard?

Neon Jian + Fei Zhua (Rec.)Zhuge repeater + Fei Zhua

Rope dartTwin butterfly swords + Fei Zhua

DECISION · 21:03

The shard's name?

Nine Dragon Stack (Rec.)Kowloon StackFog City

The Stack



mockup A · the paifang in Jiehua Neon. The bar every later frame is measured against.



clean room v1 · the first real-time three.js spawn, on the HUD



clean room v1 · the stair-street

21:06 · “loop until perfect”

NINE ANGLES, THROWAWAY LABS, BY ANY MEANS NECESSARY

- JAKE · 21:06

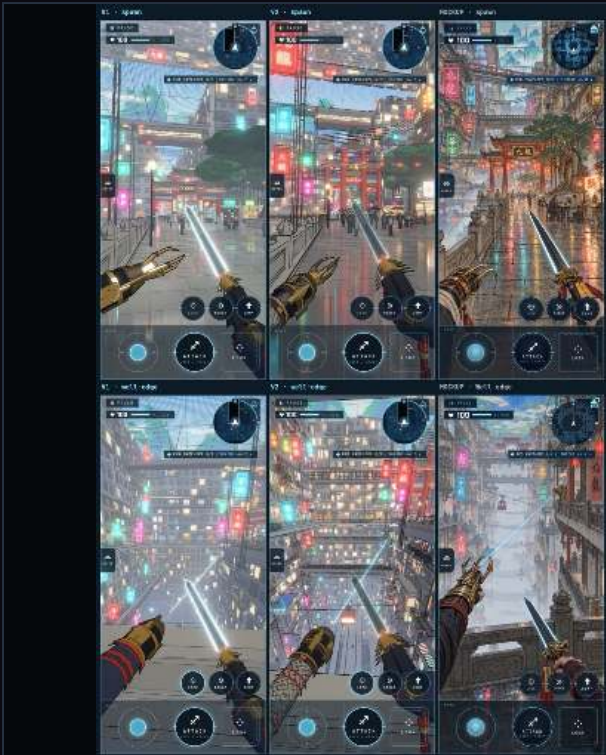
The mockups are intense. Use all the techniques available to you from all the shards and all the commit messages, whatever it takes, all the local models ... the nine-image looping with mockups and in-game images, right? That technique is not documented; fine, document that technique. You just take nine screenshots in game from the nine angles, you do a mockup of those nine angles, and you just loop until perfect. You set a really challenging art style for yourself, but I truly believe that you are actually state-of-the-art AGI Opus 5.5. So make multiple smaller clean-room prototypes to achieve a subset of the art style, and repeat with your learnings. Repeat, repeat, repeat. Whatever it takes ... just fucking get it done by any means necessary.
- JAKE · 21:07

Spawn research agents as needed to see how other games are made (AAA iOS) etc.
- JAKE · 21:09

If you say the clean room implementation is a waste of time and we should implement a fragment of the shard in game behind flags to take screenshots of it in the full build, then do that. You're approved to implement a subset of a shard that we can throw away later.

Found the technique in Pine Hollow's remaster: a look loop of 9 angles × 5 anchors (PH-0.5 / PH-L1), with in-game captures, mockups painted over those exact frames, and a LUT fitted over 54 frames. Two research agents ran (in-repo look techniques; AAA iOS stylised rendering) and pushed as 7e521fe4. Then four lab prototypes, each owning its own folder: P1 ruled-ink silk material · P2 neon, bloom, wet ground · P3 dense facade kit · P4 weapon and hero props.

Share progress photos as you get them for my enjoyment.



21:40 · neon lab final: real-time wet street | codex paint-over of the same frame | mockup

21:36 · clean room v1 → v2 → mockup. Real-time, ~2.8 ms/frame, 118 draw calls.



21:43 · facade lab: balconies, cages, laundry, pipes, lantern lines dissolving into silk fog



21:44 · ink lab: real-time (left of each pair) vs a codex Jiehua paint-over of that frame



21:53 · hero lab: the Jian + Fei Zhua, the paifang, the banyan, mahjong players (TRELLIS, then ink-shaded)

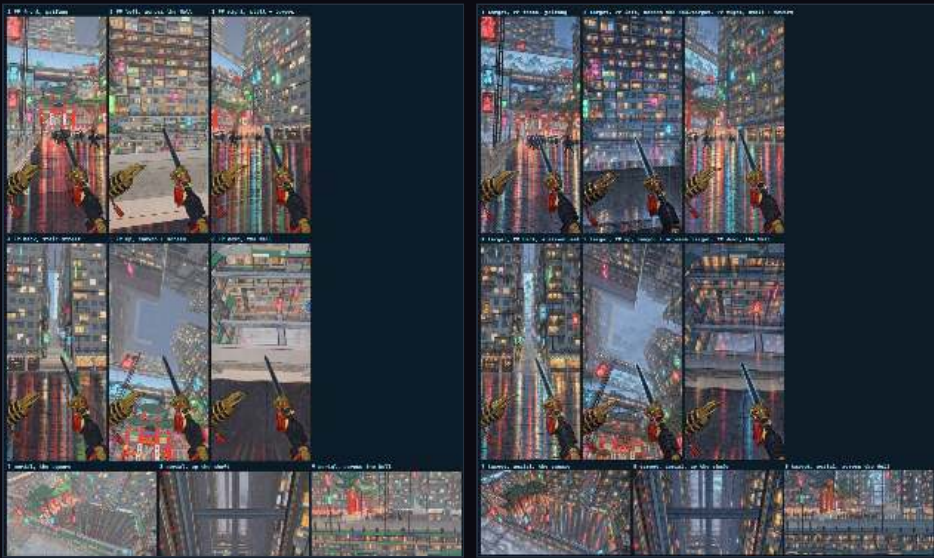


21:40 · LIVE f230117-muhs76ef: the COMING SOON card art, a clean-room capture

The COMING SOON card went live at 21:40 as f230117-muhs76ef: a slideshow backdrop of clean-room captures with a PROTOTYPE caption (tap to skip), checked on the live site at iPhone size through the browser lock.

22:03 · the look loop, then two half-domes

JAKE CORRECTS THE METHOD



Loop round 1 · the clean room with all four labs merged, from 9 frozen angles

Loop round 1 targets · codex paint-overs of those same 9 frames



blue hour vs warm silk

DECISION · 22:10

Which look should the 9-angle loop chase?

These 9 targets (Rec.)

Paler silk

Darker night

DECISION · 22:10

Default time of day for the shard's look?

A Blue hour (Rec.)

B Warm silk

JAKE · 22:10

Show me a progress image from mockup to attempt 1 to attempt n/2 to attempt current. ... What's the plan to achieve the mockups? What's the plan to get there?

The gap after v1 → v2 → now was no longer structure. Density, neon, wet streaks, rain, crowd and ink lines had closed. What was left: **surface** (the mockup's stone and wood are painted), **light** (warm pools, glowing windows, a darker value), **framing** (the weapon had drifted). Three more labs started: **P5** painted textures · **P6** light pools and grade · **P7** foliage, props, sky scroll.

JAKE · 22:14

Use as many subagents as necessary to absolutely get there. Bump the graphical quality. Obviously 30 FPS is fine on mobile; that's what we've been doing with Pine Hollow. And the more you can really polish and remaster and upscale the first person, the grappling hook, the weapon, the hands, the arms, just absolutely everything, pumped to the max.

The phone target dropped to 30 fps, which roughly doubles the triangle and texture budget. Two first-person labs: **P8** the jian, hands and arms · **P9** the whole Fei Zhua grapple.

JAKE · 22:19

Building the intense level of detail of the scene of the mockup requires something separate. The mockup is standing and looking at the gate and the tree. You need to jump forward, 30, 40, 50 metres, to just left of that tree, and then do a nine-shot mockup from that point: facing the gate, looking at the tree, looking left, looking up, looking from behind, back towards where the character in the mockup is standing. ... The nine-angle shot basically creates a half dome, and it polishes and loops and perfects that half dome. The position you're standing at now is the edge of the half dome, and you're looking towards the centre of another one. **You have to polish two half domes: the half dome of where the character is standing and the half dome of where the character is looking**, with two agents, into the same clean room. That's the only way to get where we need to get.

CLAUDE · 22:19

You're right. The loop only polishes the half-dome around where the player stands. The spawn view looks into a second half-dome around the gate and banyan, and that one never gets polished from the sides or behind. So I'll run two domes into the same clean room with two agents.



23:04 · dome B (the gate and banyan) round 1 → 4 → target: lacquer, dougong, the 九龍 plaque, carved plinths, hanging roots



23:05 · dome B round 6: the gate re-proportioned to the mockup, the crowd pushed back

JAKE · 22:46

What is image 5/5, gold on indigo? A mockup or an in-game? ... If you're going to slideshow 5 images, the hero image in the Nine Dragon Stack card on the bottom half needs to also slideshow.

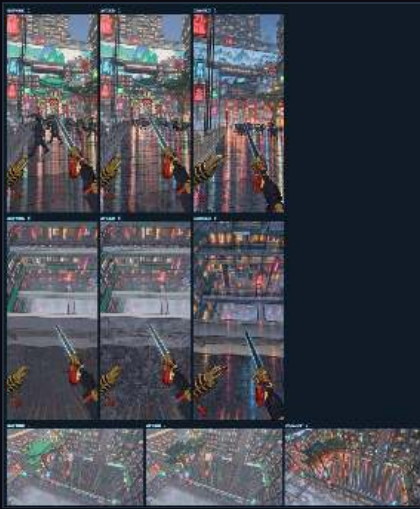
Slide 5 was in-game: the same spawn with the deep-strata style switched on. The card's own picture now crossfades in step with the backdrop and the caption. Live [da2566d-muhv80gg](#), 23:05.

JAKE · 23:14

Wait, I thought we were way further ahead. I thought we were on round six or round seven, with way more labs merged. Why does the "now" not feel as great as the previous rounds?

CLAUDE · 23:21

Round 8 went backwards. It merged the painted-texture lab: the granite made the wet ground grainy and grey and broke up the crisp neon streaks. Our colour score improved, but it can't see texture noise, so the loop counted it as a win. **Your eye was right and the metric was wrong.**



dome A round 7 board: before | after | target



round 8 - the painted-texture merge that went backwards



round 9 - the repair, judged by eye at phone size

23:23 · the first-person remaster

THE MODEL LANDED, THE MOTION DIDN'T



P8 viewmodel: the Neon Jian with a sculpted dragon guard (from a 9.4 M-face model), gloved hand, tassel and talisman, the gauntlet on a bandaged forearm



the 4x6 motion sheet: idle, walk, light slash, heavy chop, parry, draw

JAKE · 23:42

You've got a four by six of all the animations of the high-res weapons, right? And the hand rotating, the hand twisting and going in a completely broken snap fashion. Something in the animation and the rigging is completely fucked, of course. But the texture quality and the model quality is surprisingly good.

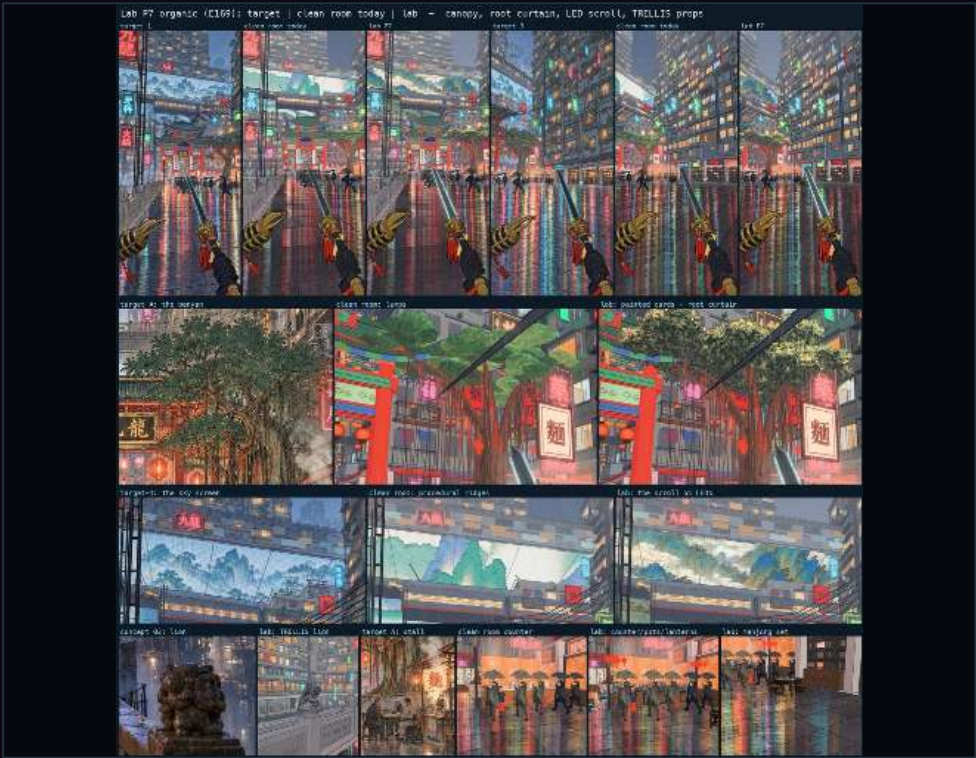
Cause: the lab animated the hand, sword and arm as separate rigid pieces with independent rotations, not as a jointed arm, and moves restarted instead of blending. The fix went back to the agent that built it. At 02:55 it came back with skinned arms, twist bones, IK from the blade's path, wrist limits and crossfades. An automated rig gate checks every frame for joint limits and snaps, and the tassel never passes through the hand.



P9 grapple: aim, lock (gold brackets), fire, the line sags, the claw bites the brass dragon hook, zip with FOV kick, vault, land, plus the miss that snaps back



the zip across the Well (a 5.7 s video was sent)



P7 organic: painted leaf canopy and root curtain, a real 千里江山图 scroll on the LED sky screens, TRELLIS stone lions



02:55 · the arm rig fix, before | after: an anatomically possible grip, no more snapping

JAKE · 23:47

There is something absolutely insane about the lighting and the reflections in the target mockup that I don't see in round nine. Lighting, reflections, post processing, ambient occlusion, basically all the fucking shit we have in the main game, all the fucking shit we have in Pine Hollow. We need all of it to get anywhere close to perfection.

23:50: usage limit. All six running agents were stopped; nothing uncommitted was deleted. Live by then: the card with the synced slideshow; committed: concept art, plan, style research, mockups on the live HUD, the look-loop write-up with the two-dome method ([docs/design/L00K-L00P.md](#)), all nine labs, dome A rounds 1–9, dome B rounds 1–8.

JAKE · 23:50

At what point do we break away from the labs and the clean room and start implementing a partial shard? There's an ungodly amount of fucking code, an ungodly amount of clean-room prototype, lab prototype, a fucking mountain of code. If we just do it in the main game instead, we utilize everything. We're doing so much fucking throwaway shit.

09-26 00:22 · break away: the partial shard in the engine

EIGHT DOMES, BUDGETS, A MILKY VEIL

JAKE · 00:22

Continue & resume; for each subagent, rescue, continue & resume it.

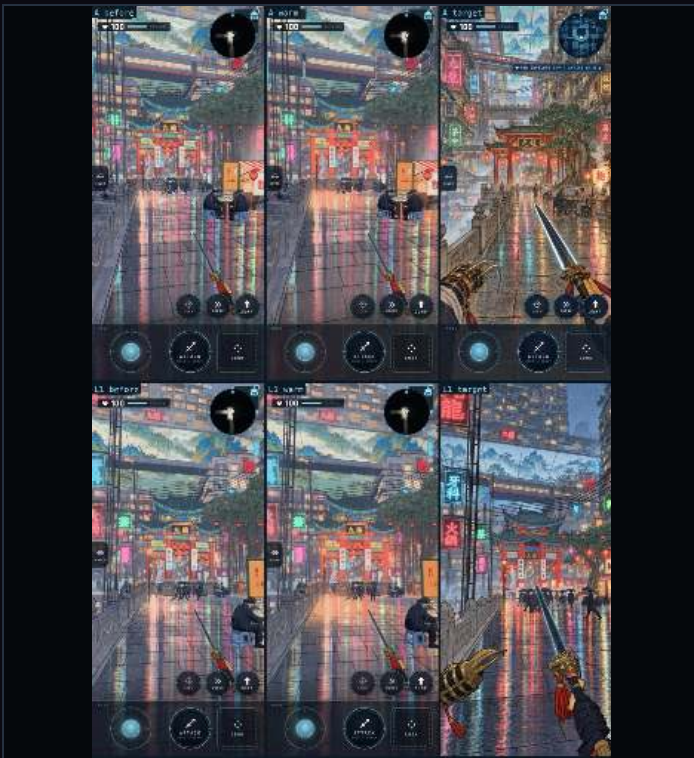
CLAUDE · 00:27

When to break away: now. The clean room did its job: it found the look, the density and the neon, and gave us the coming-soon art. Everything still missing is something the engine already has or needs anyway: real lighting, reflections, AO and post; a rigged weapon; physics; phone performance. Rebuilding those in a dev page is the throwaway work you're seeing.

JAKE · 00:29

Remember to implement a partial shard lol, not a full shard.

The rescue commit [acd28382](#) saved the stopped agents' work. Then [96d57347](#) moved the clean room's world and look into [src/chunks/nine-dragon-stack/](#) as a straight rename. A **port lead** owned the shard definition, colliders, the Debug switch, the minimap and the weapon hookup. A **render agent** owned [look/](#) and brought Pine Hollow's lighting stack across. Scope: Lantern Square, the Well's rim and upper galleries, the stair-street stub. No strata system, lifts, quest or enemies.



01:19 • in game: warm lamplight before | after | target. An amber pool on the wet stone, red gate posts, warm darks.



dome B round 8: real leaves on the banyan instead of green lumps

JAKE • 01:09

Only the first mockup is good. The other three views are pretty terrible, right? They're lacking an incredibly large amount of detail in the world modeling, and in the models placed inside the world. How do you get the world modeling in such a ridiculously vertical world to be near correct to the mockup? ... I feel like you need two domes for each mockup. That's the only fucking way it could ever work. So I need to see eight domes, the nine shots for those eight domes, a three by three I can view on my phone, and then you need to build those eight domes in engine.

JAKE • 01:13

Codex does have to somehow turn a single mockup into a 3x3, nine angles. Asking it to paint the mockups from the in-game cameras is asking it to just "improve/enhance" what we have in engine, instead of imagining the nine angles of what it sees in the original mockup. The approach seems wrong here to me.

Redirected: each dome's 3x3 is now imagined from the mockup alone. Nine camera specs plus a plain floor plan for consistency, one codex generation painting all nine views together, then each tile upscaled with the mockup as the detail reference. The corrected method went into the look-loop doc ([ba0bd590](#)).

JAKE • 01:18

Did you say there was a way for me to enter shard 4?



dome A1 - the spawn, where mockup A stands: nine angles imagined from the mockup

dome A2 - the gate and banyan, where it looks



JAKE • 01:26

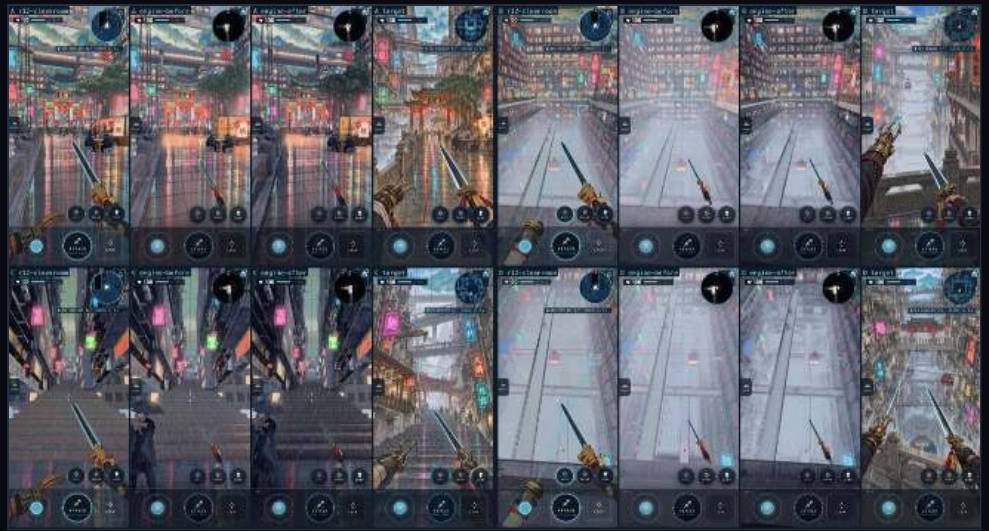
I feel like the clean room / labs / looping was producing a lot better screenshots than the in-game partial shard we're working on right now. Give me them side by side, and then ask yourself: how do we make the partial shard look as good or better than the clean-room lab screenshots?

JAKE • 01:30

Down the Well is the weakest in absolutely all of them. You're supposed to be able to look down and see a lot of detail, a lot of verticality, a lot of platforms. Looking down the Well is just flat and there's nothing. The world's made really simple, not layered and complex.



01:28 · per mockup: best clean-room frame
| in game now | mockup



01:49 · the render repair: clean room | game before | game after | target. The milky veil is gone.

The milky veil had two causes: fog measured from a stale camera position, and a work-in-progress haze value left switched on, so the image was graded twice. With both fixed, contrast and the streaks came back to the clean room's level and AO went on. Then the **triangle budget** blew: 2.82 M triangles and 168 draws at the spawn against a 2.5 M cap. Fixes: a triangle freeze for all six domes, lantern LOD (about 1,100 lanterns at full detail, saving ~390k triangles), and crowd cells of about 40 m (the crowd had been one 765k-triangle mesh that was never culled). The frame came down to 1.07–1.39 M.



02:41 · in game, on a bridge halfway down the Yamen Well looking north: stacked offset galleries, a covered bridge, the red timber bridge, hammock nets



dome D2 round 5: no floor any more, galleries all the way down into the mist



dome C1 round 2 · the stair-street vs its target

Shipped behind a Debug switch, three times: **3719d1d** (01:43, the partial shard), **f6c6276-mui2laym** (02:30, the eight-dome build so far, worst frame 2.08 M triangles), and checkpoints 2 and 3 (**67093502**, **7a339ed2**: culling from the final camera, reflections, walkable bridges with 0 stuck on six routes, and the rigged arms with their own 70° projection). 03:35: usage limit again.

06:57 · plan the rest, re-shoot the card, a trailer

THEN A WEEKLY LIMIT

JAKE · 06:57

Can you write the remainder of the shard implementation into the mega plan? ... You need to update the hero images in the carousel for coming soon too.



07:02 · the new carousel art, re-shot in the engine: the stair-street



the Well bridge, layered all the way down



the spawn in the engine with the rigged Neon Jian (as live on 09-28)

JAKE · 07:04

Can you make a 15 s trailer for this coming soon shard. I played the demo and it's very partial and it's easy to jump out of bounds lol. Not sure what cool content trailer things you can set up, maybe the grappling hook, maybe something else. It's really a partial shard for the mockups lol. Not sure how that translates to a trailer but I believe in you.

JAKE · 07:06

Plan first then trailer.

The rest of the shard went into [docs/plans/NINE-DRAGON-STACK.md \(f3e7e849\)](#): where it stands, the build method (mockups → two domes per mockup → 3×3 targets imagined from the mockup → eye-judged rounds → gated checkpoint), and the F rows for finishing the fragment. The card and carousel were live as [903d66e-muichjhc](#).

JAKE · 07:24

Shortest path to trailer.

JAKE · 07:25

Grapple is amazing so if we can do it quickly it will be the wow for the trailer.

JAKE · 07:30

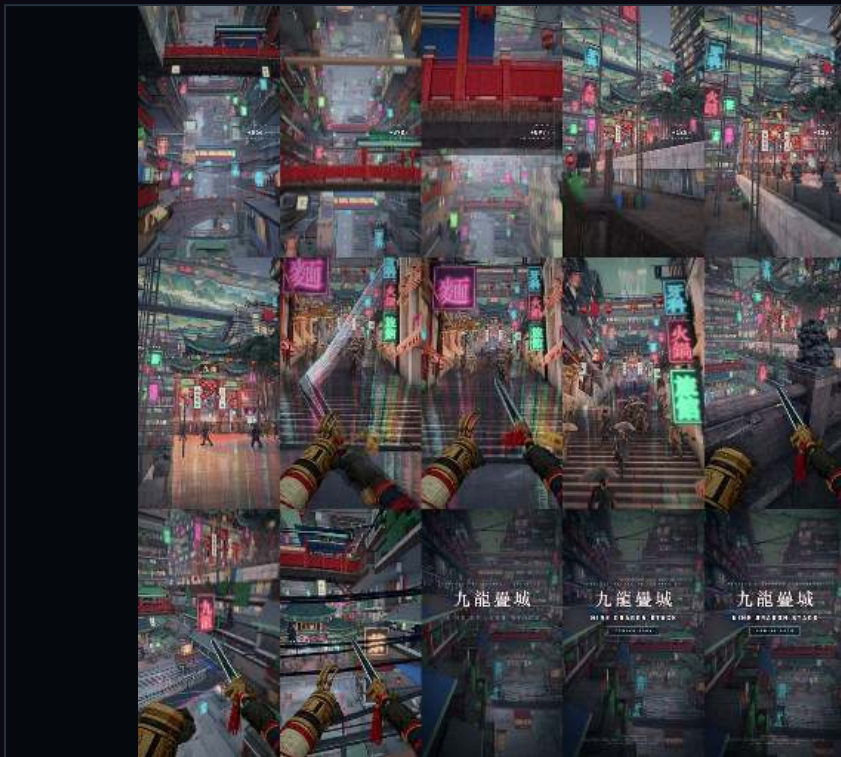
Is the sword and grapple significantly larger in the clean room implementation? Is it larger in the lab? From the images it just looks like it's all smaller in game. Something didn't translate. Can you give me the A/B comparison photo?

07:38: weekly limit, resets Sep 29. The session stopped mid-answer. The A/B photo came 2½ days later.

09-26 → 09-28 · the gap

GPT 6 SOL ON MAIN, AND THE MULTI-DRAW THAT KILLED THE IPHONE

This chapter is not from the session transcript. It comes from git and the repo's own records, because the session resumed on top of it. While this session was paused, another model (GPT 6 sol) landed **143 commits** on main. Those that touch Nine Dragon: the Fei Zhua grapple ported into the fragment, the dragon-hook sculpt placed, the 15 s portrait trailer cut and then re-cut with the live grapple ([d76d79c0](#), 09-26 18:39), phone boot-memory cuts, deferred audio and textures, and two new plan files.



the 15 s portrait trailer (round 22), contact sheet: 1080×1920, 60 fps



the grapple beat at 9.00-11.67 s: fire, the filament across the Well, the pull, the settle

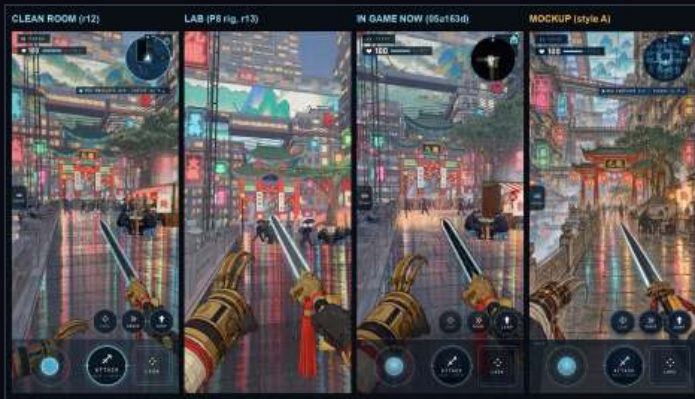
The crash. [51cf6c83](#) (09-26 18:08), "Batch Nine Dragon facade details when multi-draw is available", swapped per-piece **InstancedMesh** facades for a **BatchedMesh** over **WEBGL_multi_draw**. Fewer draw calls. On Jake's physical iPhone 17 Pro the page flashed white and reloaded: native logs show WebContent killed at 4.68-5.86 GB in Safari, and over the 2,048 MB ActiveHard limit in Chrome. The **iOS Simulator** stayed under 1 GB and never reproduced it. The isolation went through E257, E261, E264 and E265 (the paper lanterns were suspected first). The instanced build loaded and ran World Explorer for 90 s. On 09-28 Jake made it global: **E271** records the incident permanently ([docs/audits/nine-dragon-mobile-multidraw.md](#), linked from [AGENTS.md](#)), and **E272** removes facade multi-draw on every shard, tier and platform, with a regression test that requires instanced facades.

09-28 23:11 · resume: the mockup pass (E281)

FIVE LANES, ROUNDS 1 AND 2

JAKE · 23:11

Hi, can you resume / rescue your work and all the subagents' work. Before you do that, can you check on the main branch all the work GPT 6 sol did in your absence; more work was done on this branch, including making new plan files.



23:16 · the sword-size A/B: clean room r12 | lab rig r13 | in game | mockup. The same blade. The HUD deck hides the gauntlet and tassels.



23:18 · the four mockup cameras vs live 05a163d: A close; B and D have a grey rail and no misty drop; C cluttered

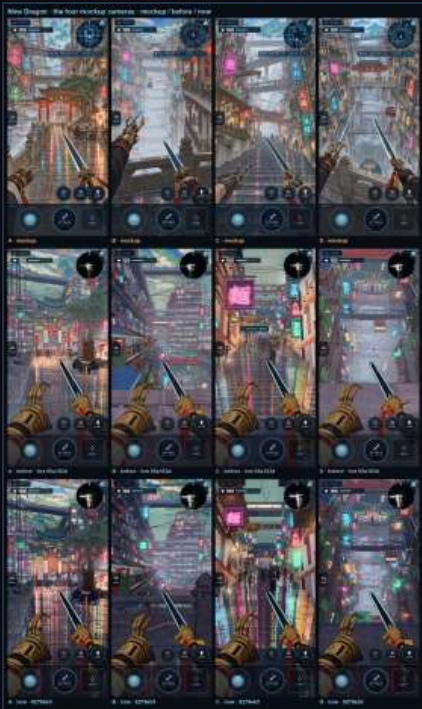
JAKE · 23:33

If you want to do a full pass at making the world look like the mockups ... I think we have four mockup images and the dome 3x3 boards. You can make multiple passes at making the game world look like the mockup. We can definitely improve the graphical fidelity and the level of detail that exists everywhere.

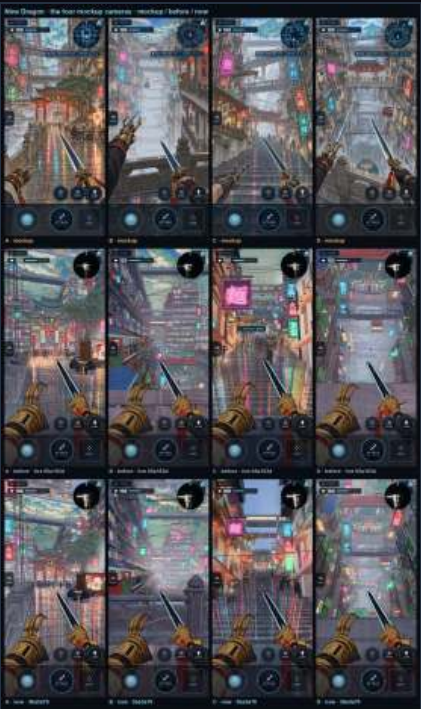
A new capture tool, [scripts/nine-dragon-domes.mjs](#), shoots each dome's nine views in the game beside its targets, plus the four mockup cameras with the HUD, and records draws, triangles and memory. Five lanes, each owning its own files and doing at least three passes: **render** (dusk grade, glow, wet) · **square + gate** (A1/A2) · **stair-street** (C1/C2) · **city fabric** (facades, lanterns, signs) · **the Well rim** (B/D). Each pass is judged at phone size against the last one and the target, and only kept if it gets closer.



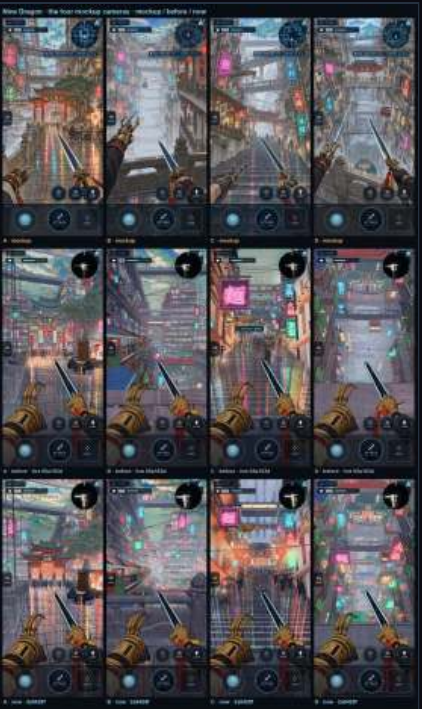
pass 0 · dome A1, target | game for nine views. The game reads as flat grey daylight; every target is blue-hour dusk. after round 1 (d6ca7e6) · the same nine views



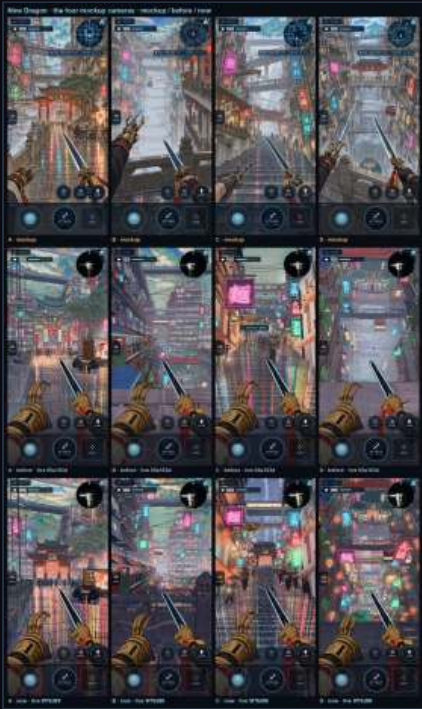
00:21 · board 1 (8279eb3): C gained most; A and C went too dark



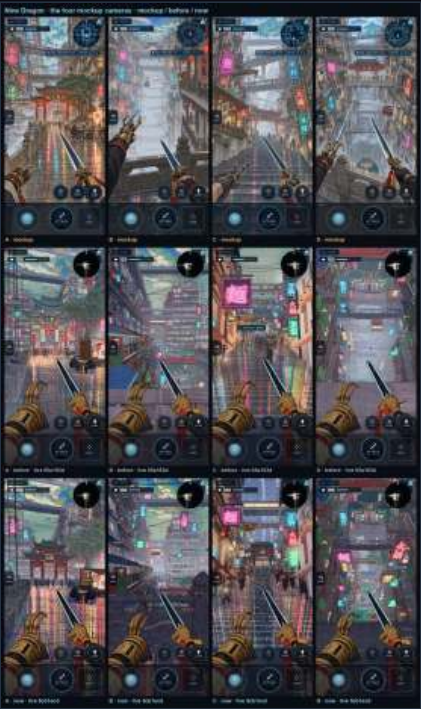
00:46 · board 2 (3ba3a70): the brightness matches; the paifang lost its lanterns



01:19 · board 3 (2a5428f): the lantern-lit gate and the sign stack are back



02:39 · round 2 live (9f76d99): B finally recedes down a 190 m lantern-lit canyon



02:48 · with the glow fix, live 5cb1ecd: D crisp again



D: the temple on a dark rock island in the cloud sea, gallery rows stepping down to it

DECISION · 06:28

Rounds 1–2 are live (5cb1ecd). What should the mockup pass do next?

Round 3 (Rec.)

Phone playtest first

Hero 3D assets

Stop here

09-29 06:35 · “we have a performance problem”

16 FPS ON THE PHONE, 30 IN THE END

JAKE · 06:35

We have a performance problem.



Jake's iPhone, 06:35 · build 5cb1ecd: 16 fps, frame p50 63 ms / p95 81 ms



08:07 · after the streak cuts (8c62342): 22 fps

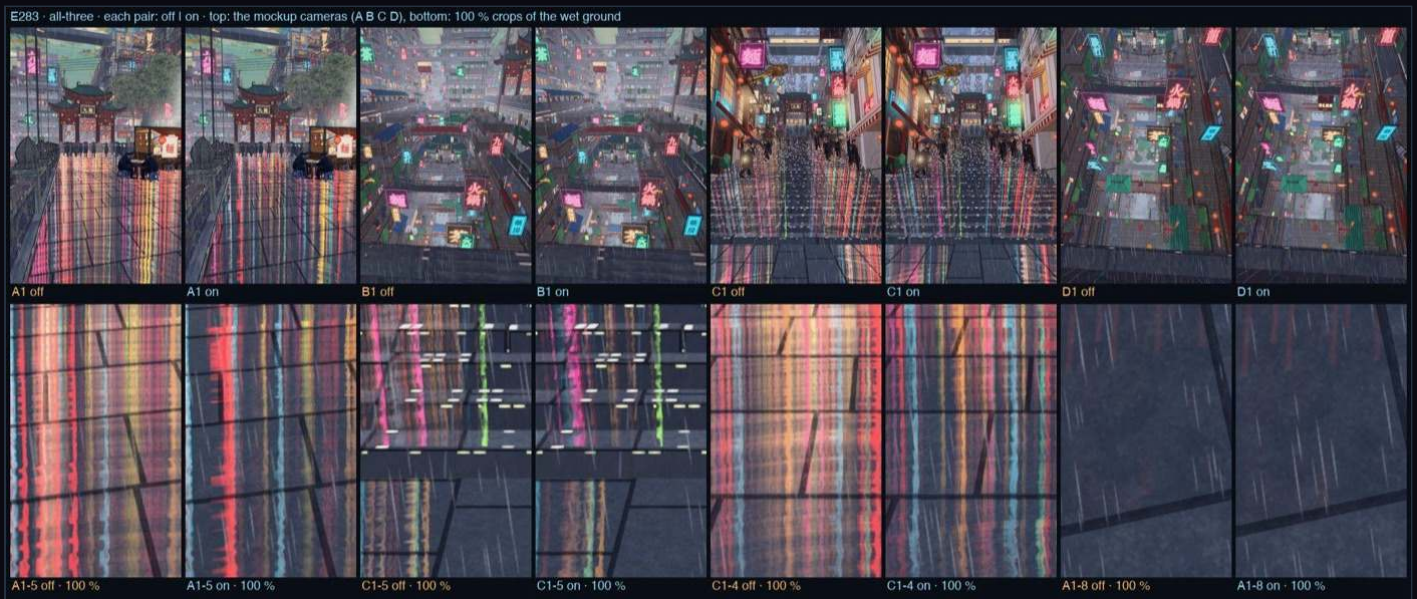


09:30 00:56 · hot, after all the cuts: "Play test on fps is not bad"

JAKE · 06:44

The baseline fps is also the wrong thing to measure, right? Sure we have a regression, but I'm sure the baseline is also not a stable 30 fps.

Root cause: the wet-ground streak cards, 1.34 ms at the spawn, 43 % of the frame; at the worst view (C1-4, the square from the stair foot) they were 57 %. The rule for every cut: if the frame stays pixel-identical it ships; if it changes the look it becomes a default-off Debug ▶ Performance row with before/after stills at the four mockup views and 100 % crops. Jake judges pictures and never toggles anything on the phone.



the three streak cuts, off | on, with 100 % crops of the wet ground. 1.6–2.0 ms → 0.11 ms at the spawn.

DECISION · 07:37

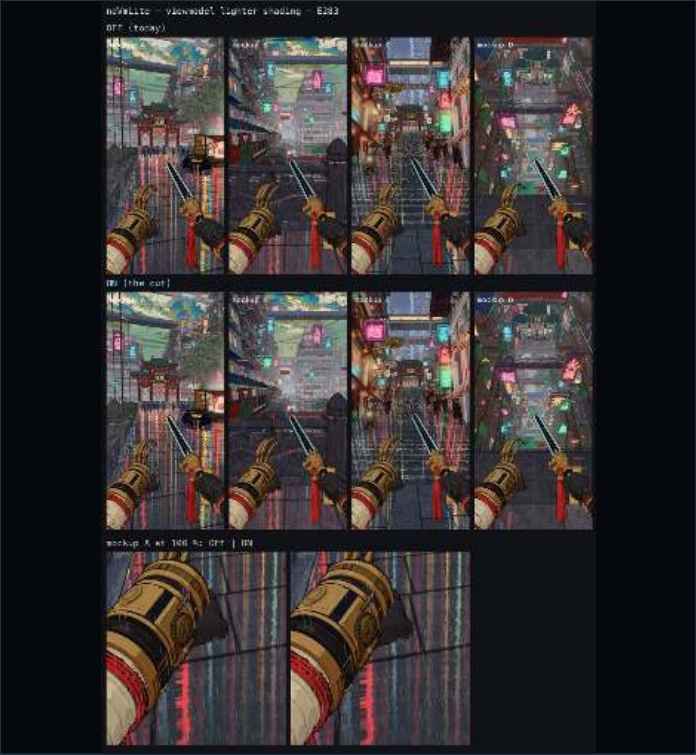
The three wet-streak cuts: make them the default?

- All three on (Rec.)
- Two on, keep the split
- Keep today's look

DECISION · 08:57

Which perf cuts should be the default?

- Lighter arm shading (Rec.)
- Stone & wall detail: none
- Stone & wall detail: near only
- Build distance LODs next



lighter arm shading: identical even at 100 %



stone & wall detail off: visible at 100 % on the balustrade posts, faint at phone size

JAKE · 08:08

Second question: the grappling hook. I can't seem to use a grappling hook. I don't think there's a dedicated HUD button for it.

DECISION · 08:12

How should the Fei Zhua grapple be triggered and shown on the phone?

LOCK relabels + hooks (Rec.)

Dedicated GRAPPLE button

One-tap JUMP

DECISION · 11:04

The two distance-LOD cuts: which should be the default?

Coarser far meshes (Rec.)

Thin far detail

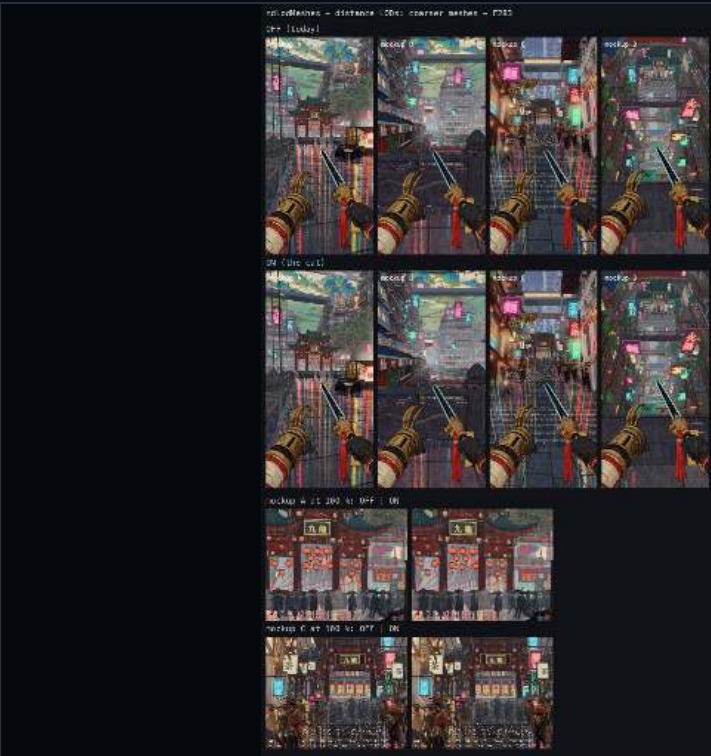
Neither

JAKE · 08:12

Then we just need more feedback about the lock button. The lock button looks disabled. Maybe change the text of the lock button to the word grapple when you're aiming correctly, and rename the jump button, once you're locked onto a grapple, from jump to zip. Those HUD buttons, we can definitely reuse them. But you just gotta make it more obvious.



09:01 · live 7c36406: LOCK lit at rest → GRAPPLE (gold ring) on a hook → LOCKED, with JUMP as ZIP. LOCK had looked disabled because it drew at 55-60 % opacity with no enemy to lock, and Nine Dragon has none.



distance LODs off | on: coarser far hooks, lions, crowd, lanterns. No visible difference.

DECISION · 11:04

What next for the frame rate?

I playtest first (Rec.)

Keep cutting, look-changing

DECISION · 09-30 00:59

Close the Nine Dragon frame-rate work (E283)?

Close it, 30 holds (Rec.)

One more reading first

Chase 60 fps next

Measured on the Mac at the phone's resolution, the spawn went from ~3.2 to 1.97 ms after the streak cuts alone. Build 5e7a13a-mumvzn1c (11:16) carried everything: the streaks, lighter arm shading, near-only stone detail, the culler and fog fixes, and both LODs, pixel-identical to the switches-on frames. Jake played it hot for 3+ minutes: 16 → 22 → 30 fps. E283 closed.

17:50 · “what is a model?”

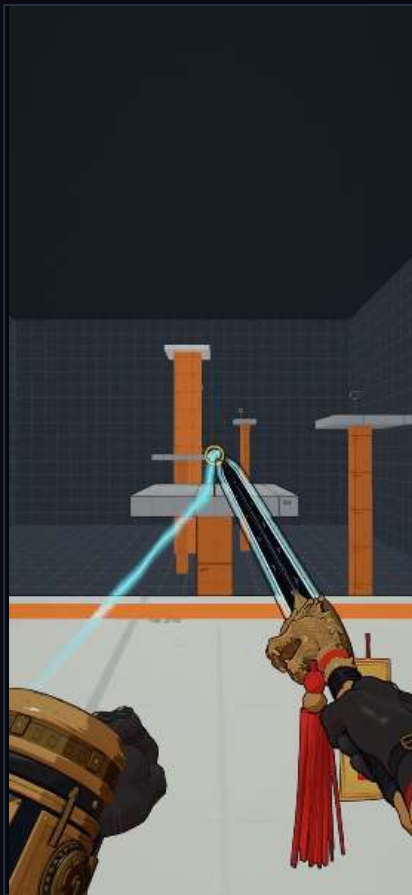
PLAYGROUNDS, A MODELS AUDIT, AN AMERICAS-ONLY RULE

JAKE · 17:50

I need to give a lot of feedback based upon the World Explorer and the Model Explorer, but I don't understand the Model Explorer. What is a model? ... Each shard has a different definition of what is significant enough to add to the Model Explorer and what is not, and some things are just baked into the world. There's nothing obvious or rigorous about it, and every agent makes an arbitrary decision about what gets registered as a model. ... Maybe we should have directories: models, a world, and a world should be made out of the models. I've heard of this word instancing, I don't know what that means. ... I think unique gameplay features need their own arenas. For Nine Dragon we want to add the grappling hook playground, just a really simple developer level, like those maps in Counter-Strike Source with all the developer textures, a custom parkour level to get a feel for the grappling hook. And Nalati has the horse riding.



19:19 · Live 2b64d80: the Explore list on Nine Dragon and Nalati, the grapple course, the horse track



grapple playground: START to FINISH in 21 s (video)



horse playground: mount, ride to half way

JAKE · 18:01

In AGENTS.md it says this game is North and South America only, and thus the licence has no implication on this game. So make sure to update everything in the repository to say that the Hunyuan 3D model is allowed.

JAKE · 18:05

Go update localai as well, and the weights project also, that Hunyuan 3D is allowed. Update those other repositories to say everything we're doing on this machine is North and South America and the limitations do not apply at all.



18:13 · the models audit: 54 Model Explorer cards against ~300 distinct shapes. Seven code paths, whole places listed as “models”, and Nine Dragon had ~60 code-built pieces with 4 cards.



19:02 · the new spin-clip tool (scripts/model-spin.mjs): Nine Dragon’s first five models, 2 s each

DECISION · 18:21

Which model architecture should we build toward (MODEL-ARCHITECTURE.md)?

B1: models per shard (Rec.)

B2: src/models//

A: relabel only

C: data-driven layouts

DECISION · 18:21

Start the first step (M0) now?

Yes, build M0 (Rec.)

Not yet

JAKE · 18:26

Yes, I want every model that is a model to be registered, moved into the correct directory, shared across all shards or per shard. The places, the zones, the sets, that can be a new thing. ... Each model should list how many times it was used in the world. How does register-and-place work with instancing and other optimizations? 30 FPS is the baseline, 60 on iOS is the ideal. Each model should say how it's made: three.js, TRELIS, Hunyuan, CC0, Blender. Figure out whether we're supposed to have .blend files or Blender scripts ... you can write lint rules, custom scripts, static analysis. An agent can make a 10-second video spinning five models in the Model Explorer that I can view on mobile in the Claude app. ... I'm happy to approve the migration on everything.

`place()` builds each model once and places it many times, and picks the technique per model: instanced (one draw for all copies), merged (copies baked into one mesh), or single. It also owns distance LODs, culling and moving colliders per placement. Before this, the shards had six LOD schemes and three cullers between them. Blender ruling: **the script is the source, the GLB is committed, a .blend never is**. A rigged .blend is 12–15 MB against a 1 MB GLB, it can't be diffed, and every author here is an agent writing code.

20:49 · the model migration (E315), all four shards

NINE DRAGON 5 → 39 CARDS, THE WORLD IDENTICAL

E300 M0 · MODEL EXPLORER BEFORE / AFTER (PHONE)



E300 M4 · NINE DRAGON MODELS (PHONE)



WILDSHARD EXPLORE MODELS WORLD

SOLID WIREFRAME FACETS PAINT TIERS



CATALOG

MARKET BOOTH (ROAST DUCKS, STEAMERS, A COOK)

src/chunks/nine-dragon-stack/models/market.ts

TRIS 7,596

DRAW CALLS 1

BUILD 0.1 ms

MADE WITH CODE

COPIES × 5

DRAWN AS instanced

Phone Budget - 0.30 % each - 1.9 % for all 5 of.

VIEW IN WORLD

20:50 · M0 on every shard: each card says how it's made (CODE · TRELLIS · HUNYUAN · BLENDER · CC0), its copies and how they're drawn. VIEW IN WORLD on Nine Dragon lands on a real lion.

22:59 · M4 Nine Dragon, 5 → 39 cards: 18 facade pieces, the guardian lion, dragon hooks, crowd, market, paper lantern ×2,583, the paifang in three variants

the M4 spin clip

Four waves in parallel (M1 Driftwood, M2 Pine Hollow, M3 Nalati + Sets, M4 Nine Dragon). Nine Dragon's frame-rate work stayed intact: at all 76 test views the draw calls, triangles, draw order and even the packed per-copy positions and colours were identical before and after. All four went live as 402a153-munn9ug3. Then M5 (creatures, people, gear), M7 (the Sets explorer) and a second pass.

JAKE · 00:50

You mentioned that we want models and sets, so therefore we need to refactor the other shards to also have models and sets. If we're going to have models and sets, then all four shards need to have models and sets.

JAKE · 00:52

Also, any questions or decisions, use the ask tool. You've just been working a lot without me. Get me back into the loop.

DECISION · 00:54

When do you want to review the Model Explorer and the new Sets explorer?

Now-ish, before M6 (Rec.)

At the very end

Per shard as it lands

DECISION · 00:54

Nine Dragon is waiting on your hot-phone playtest. Meanwhile?

Wait for my playtest (Rec.)

Prep look-changing cuts

Round 3 of the mockup pass



01:00 • Jake: "Where's the set explorer?" and "The two playgrounds in here are images."

01:26 • "The cube outline of the set ... is way too light. I can't see it."

01:28 • "Too zoomed in." The barrel fills the screen.

JAKE • 01:25

Can we make the Set Explorer render a partial world in a circle or dome around the set? I want to have a focused review. ... World Explorer is not multi-touch: I can't change speed whilst flying.

DECISION • 01:24

Nine Dragon's phone memory hasn't been measured in ~80 commits (last physical World Explorer peak was 1.054 GB, over the 1.0 GB cap). How do we check it?

Automated Simulator run (Rec.)

Simulator + your phone

Not now

DECISION • 01:24

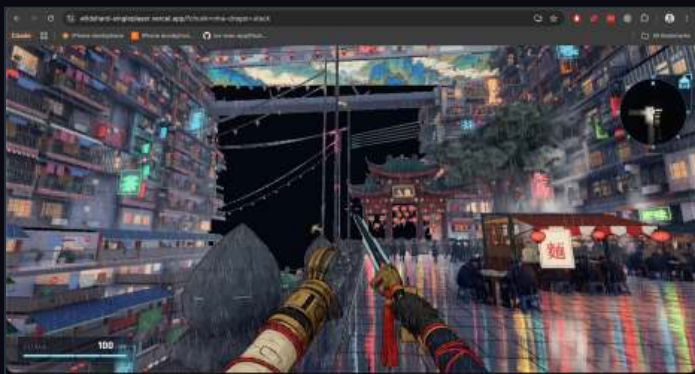
The E265 "Paper lantern geometry" Debug row was for isolating the iPhone crash (it turned out to be facade multi-draw). Remove it?

Remove it (Rec.)

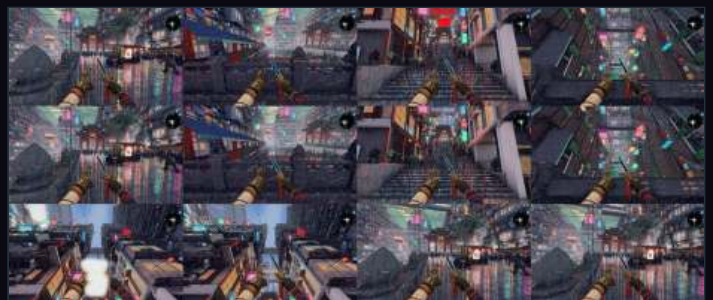
Keep it

JAKE • 01:45

The card on the Model Explorer has different heights; make it a fixed-height card. ... The card is blocking the bottom quarter of the model. The centre of the Model Explorer needs to move up: the model rendering zone is the top two thirds, and the bottom third is space for the card and the HUD.



02:09 • Jake on desktop Chrome: "black screen on desktop." The sky behind the paifang renders black.



E337 fixed (7139c2cb): the far-skyline shader squared a signed number with pow(). On Chrome for Windows that is NaN, which composites as black; Apple GPUs happen to return the right value.

Simulator memory, Nine Dragon (decimal GB, process peak / Web Inspector): loading 0.74–0.75 / 0.78–0.83 (cap 1.8); World Explorer 0.48–0.50 / 0.75 (cap 1.0). Nothing moved more than 0.06 GB through the frame-rate cuts and the migration. The Simulator isn't the phone, though: it showed numbers like these when the iPhone read 1.054.

JAKE · 03:12

I think we have too many subagents, max 3 subagents.

DECISION · 03:14

Right now 6 subagents are running. Which 3 keep going?

Sky + Explorer + 2nd pass (Rec.)

Sky + Explorer + multi-touch

Sky + Explorer + memory

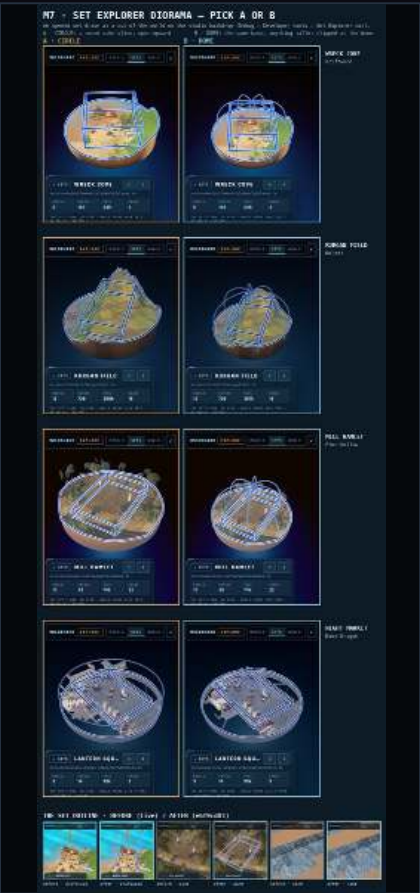
JAKE: 1, but also merge units of work into larger subagents.

JAKE · 03:15

I'm going AFK, continue autonomously.

09-30 05:44 · the last picks, the leftovers, done

“OK CAN YOU JUST DO THEM NOW LOL”



Set Explorer diorama: A circle | B dome



Nine Dragon second pass



Model Explorer framing: the model in the top two thirds, the card in the bottom third

DECISION · 05:46

Set Explorer diorama (board 1): which cut becomes the default?

A · Circle (Rec.)

B · Dome

Neither: off

DECISION · 05:46

Some Nine Dragon models open dark or small in the Model Explorer. Fix it?

Brighten + refit (Rec.)

Leave as is

DECISION · 05:46

Can M6 (deleting the old code paths) start once the banyan fix and E342 land?

Yes, start M6 (Rec.)

I'll review on my phone first

Changes first

DECISION · 05:46

After M6, what's next for this session?

E281 round 3 look pass (Rec.)

Phone memory first

Something else

JAKE: Wrap up the session after.



E342 · VIEW IN WORLD on Nine Dragon now lands with a clear view: 6/20 → 19/20 taps select their own model

Nine Dragon specimens brightened and refitted (Explorer only; the world is unchanged)

07:11: E315 done. M6 deleted the old code paths ([e96f7e22](#), [0ddfb7c3](#)) and the model check now covers every area of the code. The grey banyan turned out to be the city's fog drawn into the Model Explorer studio. The plan was archived at [project/archive/2026-09-30-model-architecture.md](#), live [24cc132-muo2dk2d](#). Meanwhile the Driftwood loot session gave Nine Dragon its Bag: MAP + GEAR, the “Neon Jian” and a Fei Zhua tool card, and no iron sword.

JAKE · 08:42
What's left and what's next?

DECISION · 08:43
What should the next session pick up?

E281 round 3 + F8 (Rec.) F3 teaser trailer Migration leftovers P1: the full shard

JAKE · 08:48
Ok can you just do them now lol.



E346 · the Well-rim balustrade, paifang posts and bridge posts are models now: 76 views identical, colliders within 1 µm, 0 stuck



E348 · dummies, hoverboard, swim hands, reins, butterfly, raptor, arrows get cards



E345 · every shard passes the tap test: Nine Dragon 20/20

Three lanes, six asks (E344–E349), all live as **855d81c-muo90usf** at 10:19 with the world unchanged. At 10:48: **3cc6729-muo9m3aa** is HEAD and live, CI green, nothing unpushed, no tracked file modified, the four gates green on a clean export. Then **/prepare-to-exit**.

DECISION · 10:47

What should this session do next?

E281 round 3 (Rec.)

E354 trail stuck points

Wrap up

Where Nine Dragon stands (plan State line, 09-30): **blocked**, paused for E357 until M1 of GAME-NORMALIZATION, then re-planned for the new engine. The fragment stays playable and live: Pause ▶ Settings ▶ Debug ▶ Nine Dragon Stack prototype, then the EXPERIMENTAL card. Next for the look: E281 round 3.

Deploys

EVERY PRODUCTION BUILD THE SESSION SAW GO LIVE

TIME	BUILD	WHAT WENT LIVE
09:25 19:50	25166fa-muho86w1	Jiehua Neon mockups on the phone HUD (the baseline for the fresh captures)
21:40	f230117-muhs76ef	the Nine Dragon Stack COMING SOON card + slideshow + PROTOTYPE caption
22:39	17f27bd-muhubywz	new card art from loop round 2
23:05	da2566d-muhv80gg	the card's own picture slideshows in step with the backdrop
09:26 01:43	3719d1d-mui0w4ju	the in-engine partial shard behind the Debug switch: square, Well rim, stair stub
02:30	f6c6276-mui2laym	checkpoint: the eight-dome build, lighting parity, layered mist, lantern LOD
03:35	7a339ed2 (pushed)	checkpoints 2-3: culling, reflections, walkable bridges, the rigged arms
07:09	903d66e-muichjhc	card + carousel re-shot from the in-engine partial shard
09:28 23:12	05a163d-muLpho5m	GPT 6 sol's state: grapple, trailer, facade multi-draw removed (E272)
09:29 01:34	d6ca7e6-mumavuoZ	mockup pass round 1 (five lanes)
02:38	9f76d99-mumd6dd3	round 2: the 190 m canyon, the temple island, the night market
02:47	5cb1ecd-mumdhfh3	round 2 glow fix (the 16 fps build)
07:47	8c62342-mumo6tgc	the three streak cuts made permanent (22 fps)
09:01	7c36406-mumqTy9w	the grapple on LOCK → GRAPPLE / JUMP → ZIP, more hooks
11:16	5e7a13a-mumvnzlc	all perf cuts + both LODs: the 30 fps build
19:19	2b64d80-muncxtha	Explore feature playgrounds: the grapple course, the horse track
20:52	be264e7-mung9cti	M0: defineModel / place(), Explorer facts on every card
09:30 00:09	402a153-munn9ug3	all four shards' model migrations (Nine Dragon 5 → 39 cards)
01:12	6ca63ec-munpj8lp	the Set Explorer
03:06	0f76317-muntlcjm	second pass: Driftwood 67 cards, Nalati 87 + 17 Sets
06:41	2e7d582-muo186ot	E342 view-in-world, the circle diorama, Nine Dragon specimens refit
07:11	24cc132-muo2dk2d	M6: old code paths deleted, E315 done and archived
10:19	855d81c-muo90usf	the six migration leftovers E344-E349
10:49	3cc6729-muo9m3aa	the wrap-up: HEAD = live, CI green

The magic

WHY THIS ONE WORKED, AND WHERE IT DIDN'T YET

- 1. **Jake fixed the method, three times.** One 9-angle dome polishes only where you stand, so use two half-domes per mockup: where you stand and where you look (22:19). Eight domes, not four (01:09). Targets imagined from the mockup, not painted over our own frames, which only “enhance” what exists (01:13). Each correction moved the frames more than any lab did.
- 2. **Throwaway labs, then a hard break.** Nine small labs found the ink, the neon streaks, the facade kit, the rig and the grapple fast, because nothing else depended on them. Then at 00:27 everything moved into the engine with one rename, since lighting, physics and phone performance only exist there.
- 3. **The eye beats the metric.** Round 8 scored better on colour and looked worse; Jake saw it from his phone. After that every pass was judged by eye at phone size against the last pass and the target.
- 4. **Pictures, not toggles.** Every perf cut that changed the look became a default-off Debug row with stills at the four mockup views and 100 % crops. Jake picked from images, and the frame rate went 16 → 22 → 30 without him flipping anything on the phone.
- 5. **Proof the rest stayed put.** The partial shard, the model migration and the leftovers each shipped with a check that the other shards were unchanged: identical frames at 76 views, colliders within 1 μm, 0 stuck in walk tests.
- 6. **The real phone is the only truth for memory.** The facade multi-draw passed desktop, the Simulator (under 1 GB) and the draw-call counts, then killed WebContent at 4.7–5.9 GB on the iPhone. It is now banned everywhere, with a written record and a regression test.

The ask ledger

DOCS/TASKS/ASKS/ – THE ONES THIS BUILD OPENED OR CLOSED

ASK	WHAT	STATUS
E169	prepare shard 4: concept art, mega plan, mockups, COMING SOON, clean room → the partial shard	open: carried by NINE-DRAGON-STACK.md (blocked on E357)
E265	triage GPT 6 sol's graphics commits as suspects for the Nine Dragon memory crashes	done: it was facade multi-draw; lantern row removed 621343eb
E271	document the facade multi-draw that killed the game, permanently	done 09-28
E272	disallow facade multi-draw across the baseline and every shard	done 09-28
E281	full pass at making the world look like the four mockups and eight domes	paused after round 2; round 3 is next
E283	“We have a performance problem”: 16 fps on the iPhone	done: 30 holds hot
E286	the grapple has no HUD button → LOCK/GRAPPLE, JUMP/ZIP	needs Jake's feel check (7c36406)
E306	what is a model? audit every way a model is made	done → E315
E307	Explore feature playgrounds: grapple course, horse track	needs Jake's feel notes (2b64d80)
E315	the whole model migration, M0–M12, four shards	done, archived 09-30
E325	real card art for the two playground cards	done
E337	black sky on desktop Chrome in Nine Dragon	done: 7139c2cb, Jake confirmed
E342	VIEW IN WORLD lands inside the stack	done: 6/20 → 19/20, then 20/20
E344–E349	the six migration leftovers (cove colliders, taps, Nine Dragon posts, homestead, live things, main.ts)	done, live 855d81c